LITERATURE REVIEW

The No-U-Turn Sampler: Adaptively Setting Path Lengths in Hamiltonian Monte Carlo

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ANALYTICAL ABSTRACT

Hamiltonian Monte Carlo (HMC) produces distant, high-acceptance proposals from the gradient of a log posterior, but its efficiency depends on a step size and a number of leapfrog steps that the user must tune. The paper introduces the No-U-Turn Sampler (NUTS), which builds a trajectory by repeatedly doubling a binary tree of leapfrog states, forwards or backwards in fictitious time, and stops when the two ends of any balanced subtree begin to approach each other. A slice variable and a careful choice of admissible states keep the target distribution invariant. The step size is tuned during burn-in by a modified Nesterov dual averaging scheme aimed at a target acceptance statistic. On a 250-dimensional correlated normal, two logistic regressions and a stochastic volatility model for S&P 500 returns, effective sample size per gradient evaluation shows NUTS matching well-tuned HMC and beating it by about a factor of three on two problems. This review reconstructs the invariance argument and the adaptation scheme, and adds a reviewer simulation on a 50-dimensional Gaussian in which NUTS matches the best HMC setting but does not exceed it. The main limitations are the absence of a formal convergence proof for the adaptation, a narrow set of benchmarks, and silence on funnel-shaped and multimodal posteriors.

1 Introduction

1.1 Paper identification and classification

The paper is a methodological and computational contribution to Bayesian statistics. Its field is Bayesian computation, and its subfield is gradient-based Markov chain Monte Carlo (MCMC) for continuous parameters. The version read in full is the arXiv preprint 1111.4246, retrieved as a complete text rendering, including the algorithms, equations, all experiments, the appendix on effective sample size and the bibliography. It describes the Stan system as forthcoming, so it predates the 2014 journal version in the Journal of Machine Learning Research; the figures were not available in the text rendering, so figure content is taken from the captions and from the authors' prose. No PDF of the paper could be downloaded in the review environment, so no copy is stored beside this review.

1.2 Executive understanding

WHAT IS THE PAPER TRYING TO DISCOVER?

The paper does not test an economic hypothesis. It asks whether Hamiltonian Monte Carlo can be made to work without the user choosing the number of leapfrog steps L and the step size ε, while keeping the target distribution exactly invariant and the efficiency of a hand-tuned sampler.

WHY DOES THE QUESTION MATTER?

Efficient posterior simulation for hierarchical and high-dimensional models is the bottleneck of applied Bayesian analysis, including Bayesian volatility, term structure and asset pricing models in finance. HMC scales roughly as $O(D^{5/4})$ per independent sample in dimension D, against $O(D^2)$ for random-walk Metropolis (Creutz, 1988, as cited by the authors), but only if ε and L are right. A sampler that tunes itself is the precondition for a general-purpose inference engine.

WHAT WAS KNOWN BEFORE, AND WHAT IS MISSING?

Neal (2011) had shown that HMC suppresses random-walk behaviour, and the adaptive MCMC literature (Andrieu and Thoms, 2008) offered Robbins and Monro (1951) schemes for tuning scalars such as ε. What was missing was a principled, automatic rule for the trajectory length L that preserves detailed balance, and a faster-forgetting adaptation of ε than the diminishing steps of Robbins and Monro.

WHAT DOES THE PAPER PROPOSE, HOW DOES IT INVESTIGATE, AND WHAT DOES IT FIND?

It proposes NUTS for L and a dual averaging scheme for ε. It investigates them by running both NUTS and tuned HMC on four targets and measuring effective sample size (ESS) per gradient evaluation. It reports that NUTS is at least as efficient as the best HMC trajectory length on every target, about

three times better on two, and that the best HMC trajectory length varies by roughly a factor of 100 across targets.

2 Research Problem and Hypothesis

2.1 Research question

PAPER Can a Hamiltonian Monte Carlo sampler choose its own trajectory length per iteration, by a stopping rule based on whether the simulated particle has started to double back, in a way that leaves the target invariant, and can the step size be tuned online so that the combined sampler needs no user input and is no less efficient, per gradient evaluation, than a well-tuned HMC?

2.2 Hypotheses

- H1 (stated). A criterion based on the sign of the inner product between the displacement from the start and the current momentum identifies when a longer trajectory no longer increases distance from the start, and can be made reversible by a doubling procedure.
- H2 (stated). Dual averaging applied to $\log \varepsilon$ drives a realised acceptance statistic to a chosen target δ.
- H3 (stated). The resulting NUTS performs at least as efficiently as the best fixed-L HMC, ignoring the cost of tuning L.
- H4 (implicit). A single default target ($\delta \approx 0.6$ for NUTS, 0.65 for HMC) is adequate across problems.

2.3 Existing literature and research gap

PAPER The authors position NUTS against Gibbs sampling (Geman and Geman, 1984), random-walk Metropolis (Metropolis et al., 1953), HMC (Duane et al., 1987; Neal, 2011), slice sampling with doubling (Neal, 2003) and adaptive MCMC (Andrieu and Thoms, 2008). They state that setting L typically requires costly preliminary runs and expertise.

CRITIQUE The stated gap is genuine and the framing is fair. The review of competing gradient-based methods is thin: Riemannian manifold HMC (Girolami and Calderhead, 2011) is mentioned only in the discussion, and the windowed HMC of Neal (1994) only as a possible explanation of the performance gains.

2.4 Claimed contribution

CRITIQUE Two contributions are claimed: NUTS and the dual averaging step size scheme. The first is demonstrated on the metric chosen, ESS per gradient, within the tested models. The second is demonstrated empirically but not proved: the convergence guarantee is borrowed from Nesterov (2009) under assumptions (monotonicity of the expected statistic in ε) that are stated but not checked for NUTS, whose statistic is not a true Metropolis acceptance probability.

3 Theoretical Framework and Mechanism

3.1 Hamiltonian Monte Carlo and the leapfrog integrator

PAPER HMC augments the parameter vector $\theta \in \mathbb{R}^D$ with a momentum $r \in \mathbb{R}^D$ and targets the joint density

$$p(\theta, r) \;\propto\; \exp\Big\{\mathcal{L}(\theta) - \tfrac{1}{2}\, r \cdot r\Big\}, \tag{1}$$

$\mathcal{L}(\theta)$	Log of the (unnormalised) target density; a negative potential energy.
$\frac{1}{2} r \cdot r$	Kinetic energy with unit mass; the momenta are independent standard normal.
$\log p(\theta, r)$	The negative total energy (Hamiltonian) of the fictitious particle.

Because r is independent of θ, the marginal of θ is the target. Hamiltonian dynamics are simulated by the leapfrog map

$$r_{t+\varepsilon/2} = r_t + \tfrac{\varepsilon}{2} \nabla_\theta \mathcal{L}(\theta_t), \qquad \theta_{t+\varepsilon} = \theta_t + \varepsilon\, r_{t+\varepsilon/2}, \qquad r_{t+\varepsilon} = r_{t+\varepsilon/2} + \tfrac{\varepsilon}{2} \nabla_\theta \mathcal{L}(\theta_{t+\varepsilon}). \tag{2}$$

Each sub-update changes either r as a function of θ or θ as a function of r, so each is a shear with a unit Jacobian determinant and the composition preserves volume. It is also reversible once the momentum is negated. After L steps the proposal $(\tilde{\theta}, \tilde{r})$ is accepted with probability

$$\alpha = \min\left\{1,\; \frac{\exp\{\mathcal{L}(\tilde{\theta}) - \tfrac{1}{2}\tilde{r} \cdot \tilde{r}\}}{\exp\{\mathcal{L}(\theta_{m-1}) - \tfrac{1}{2} r_0 \cdot r_0\}}\right\}. \tag{3}$$

Volume preservation is what lets the acceptance ratio omit a Jacobian term. If the dynamics were exact, energy would be conserved and $\alpha = 1$; the discretisation error in energy is proportional to ε^2 for large L (Leimkuhler and Reich, 2004, as cited).

EVIDENCE A worked case shows both the accuracy and the stability limit of the integrator. For a one-dimensional standard normal, $\mathcal{L} = -\theta^2/2$ and $\nabla \mathcal{L} = -\theta$, and one leapfrog step is the linear map

$$\begin{pmatrix} \theta' \\ r' \end{pmatrix} = \begin{pmatrix} 1 - \varepsilon^2/2 & \varepsilon \\ -\varepsilon(1 - \varepsilon^2/4) & 1 - \varepsilon^2/2 \end{pmatrix} \begin{pmatrix} \theta \\ r \end{pmatrix}.$$

Its determinant is $(1-\varepsilon^2/2)^2+\varepsilon^2(1-\varepsilon^2/4)=1$, confirming volume preservation, and its trace is $2-\varepsilon^2$. The eigenvalues are complex of unit modulus, so the orbit is a stable rotation, if and only if $|2-\varepsilon^2|<2$, that is $\varepsilon<2$. For a multivariate normal with precision matrix A the condition is $\varepsilon<2/\sqrt{\lambda_{\max}(A)}$. In the reviewer simulation below, $\lambda_{\max}(A)=2.91$ gives a ceiling of 1.17, and the adapted NUTS step size of about 0.73 lies below it. This is the origin of the paper's statement that an ε that is too large gives low acceptance rates.

3.2 The U-turn criterion

PAPER The time derivative of half the squared distance from the starting position θ to the current position $\tilde{\theta}$ is

$$\frac{d}{dt}\frac{(\tilde{\theta}-\theta)\cdot(\tilde{\theta}-\theta)}{2}=(\tilde{\theta}-\theta)\cdot\tilde{r}. \tag{4}$$

$\tilde{\theta}-\theta$ Vector from the starting position to the current position.

$\tilde{r}$ Current momentum, which equals $d\tilde{\theta}/dt$ with unit mass.

When the right-hand side turns negative, a further infinitesimal move would bring the particle closer to its start. Stopping at the first such time would be natural, but the stopping time then depends on the starting point in a way that breaks time reversibility. The remedy is a doubling procedure borrowed from slice sampling (Neal, 2003).

3.3 Augmented model and conditions C.1 to C.4

PAPER A slice variable u and two finite sets of states, $\mathcal{B}\supseteq\mathcal{C}$, are added. The joint density is

$$p(\theta,r,u)\propto\mathbb{I}\Big[u\in\big[0,\exp\{\mathcal{L}(\theta)-\tfrac{1}{2}r\cdot r\}\big]\Big], \tag{5}$$

whose marginal over u recovers Equation (1). Here $\mathcal{B}$ is every state the leapfrog integrator visits in one iteration and $\mathcal{C}$ is the subset to which a move is allowed. The random construction of $(\mathcal{B},\mathcal{C})$ must satisfy:

- C.1: every element of $\mathcal{C}$ is reached by volume-preserving maps, so the density can be treated as a mass function on $\mathcal{C}$.
- C.2: the current state belongs to $\mathcal{C}$ with probability one.
- C.3: every state in $\mathcal{C}$ lies in the slice, $u\le\exp\{\mathcal{L}(\theta')-\tfrac{1}{2}r'\cdot r'\}$.
- C.4: for any two states in $\mathcal{C}$, the probability of generating the pair $(\mathcal{B},\mathcal{C})$ is the same.

Given these, one iteration is: (1) draw $r\sim\mathcal{N}(0,I)$; (2) draw u uniformly on $[0,\exp\{\mathcal{L}(\theta_t)-\tfrac{1}{2}r\cdot r\}]$; (3) draw $(\mathcal{B},\mathcal{C})$ from $p(\mathcal{B},\mathcal{C}\mid\theta_t,r,u,\varepsilon)$; (4) move to a new (θ_{t+1},r) by a kernel T that leaves the uniform

distribution on $\mathcal{C}$ invariant, that is

$$\frac{1}{|\mathcal{C}|}\sum_{(\theta,r)\in\mathcal{C}} T(\theta', r' \mid \theta, r, \mathcal{C}) = \frac{\mathbb{I}[(\theta', r') \in \mathcal{C}]}{|\mathcal{C}|}. \tag{6}$$

Steps 1 to 3 together are a valid Gibbs update. Step 4 is valid because, by C.1 to C.4,

$$p(\theta, r \mid u, \mathcal{B}, \mathcal{C}, \varepsilon) \propto p(\mathcal{B}, \mathcal{C} \mid \theta, r, u, \varepsilon)\, \mathbb{I}\left[u \le e^{\mathcal{L}(\theta)-\frac{1}{2}r\cdot r}\right] \propto \mathbb{I}[(\theta, r) \in \mathcal{C}], \tag{7}$$

so the conditional of the state given the sets is uniform on $\mathcal{C}$ and any kernel preserving the uniform law on $\mathcal{C}$ preserves the augmented target.

3.4 Tree building, stopping and the admissible set

PAPER The set $\mathcal{B}$ is generated by doubling. At height j a direction $v_j \sim \text{Uniform}\{-1, 1\}$ is drawn and 2^j leapfrog steps of size $v_j\varepsilon$ are taken from the appropriate end of the existing trajectory, creating a balanced binary tree whose leaves are states. Given the final tree and a start state, at most one sequence of directions reproduces it, which delivers C.4 once inadmissible states are excluded: a state started from which the stopping rule would fire before the full tree is completed could not have generated $\mathcal{B}$.

Two stopping rules apply. The first guards against divergent simulation, with $\Delta_{\max}$ a large constant (1000 is recommended):

$$\mathcal{L}(\theta) - \tfrac{1}{2}r\cdot r - \log u < -\Delta_{\max}. \tag{8}$$

The second is the U-turn rule applied to every balanced subtree of height at least one, using its leftmost and rightmost leaves (θ^-, r^-) and (θ^+, r^+):

$$(\theta^+ - \theta^-)\cdot r^- < 0 \quad \text{or} \quad (\theta^+ - \theta^-)\cdot r^+ < 0. \tag{9}$$

Equation (9) says that continuing the dynamics forwards from θ^+ or backwards from θ^- would shrink the span of the subtree. Checking it for all $2^j - 1$ subtrees of a height-j tree costs $2^{j+1} - 2$ inner products, comparable to the number of leapfrog steps and negligible against gradient cost except for trivial models. The admissible set $\mathcal{C}$ is built as follows: include the starting state; keep only states in the slice (C.3); and if the doubling stopped because a rule fired inside the last doubling, discard every state added in that last doubling, since starting from any of them the rule would have fired earlier than for the true tree. If instead the rule fired only at the root of the full tree, no discarding is needed.

3.5 Efficient NUTS and the transition kernel

PAPER Sampling uniformly from $\mathcal{C}$ makes small jumps and needs $O(2^j)$ memory. The paper instead uses, with $w = (\theta, r)$ and $\mathcal{C} = \mathcal{C}_{\text{old}} \cup \mathcal{C}_{\text{new}}$ disjoint,

$$T(w' \mid w, \mathcal{C}) = \begin{cases} \dfrac{\mathbb{I}[w' \in \mathcal{C}_{\text{new}}]}{|\mathcal{C}_{\text{new}}|} & |\mathcal{C}_{\text{new}}| > |\mathcal{C}_{\text{old}}|, \\ \dfrac{|\mathcal{C}_{\text{new}}|}{|\mathcal{C}_{\text{old}}|} \dfrac{\mathbb{I}[w' \in \mathcal{C}_{\text{new}}]}{|\mathcal{C}_{\text{new}}|} + \left(1 - \dfrac{|\mathcal{C}_{\text{new}}|}{|\mathcal{C}_{\text{old}}|}\right) \mathbb{I}[w' = w] & |\mathcal{C}_{\text{new}}| \le |\mathcal{C}_{\text{old}}|, \end{cases} \tag{10}$$

for $w \in \mathcal{C}_{\text{old}}$. In words, propose a uniform draw from the newly added half-tree and accept with probability $\min\{1, |\mathcal{C}_{\text{new}}|/|\mathcal{C}_{\text{old}}|\}$. This is a Metropolis-Hastings kernel that satisfies detailed balance with respect to the uniform law on $\mathcal{C}$:

$$p(w \mid \mathcal{C})\, T(w' \mid w, \mathcal{C}) = p(w' \mid \mathcal{C})\, T(w \mid w', \mathcal{C}). \tag{11}$$

INFERENCE To see (11) in the case $|\mathcal{C}_{\text{new}}| \le |\mathcal{C}_{\text{old}}|$, take $w \in \mathcal{C}_{\text{old}}$ and $w' \in \mathcal{C}_{\text{new}}$. Forward flow is $\frac{1}{|\mathcal{C}|} \cdot \frac{|\mathcal{C}_{\text{new}}|}{|\mathcal{C}_{\text{old}}|} \cdot \frac{1}{|\mathcal{C}_{\text{new}}|} = \frac{1}{|\mathcal{C}||\mathcal{C}_{\text{old}}|}$. The reverse move, regarded with the roles of the two halves exchanged, has proposal probability $1/|\mathcal{C}_{\text{old}}|$ and acceptance probability one because the destination half is the larger, giving the same value. Applying T after every doubling, each time to the newest half, increases the chance of a long jump, in the manner of delayed rejection (Tierney and Mira, 1999, as cited), without paying for evaluating extra proposals. The sample from each subtree is built recursively: if two child subtrees contribute n' and n'' admissible states, the representative of the parent is the child's representative with probabilities $n'/(n'+n'')$ and $n''/(n'+n'')$, reflecting the factorisation

$$p(\theta, r \mid \mathcal{C}') = \frac{1}{|\mathcal{C}'|} = \frac{|\mathcal{C}_{\text{subtree}}|}{|\mathcal{C}'|} \cdot \frac{1}{|\mathcal{C}_{\text{subtree}}|}. \tag{12}$$

Only $O(j)$ states are stored. Algorithm 3 of the paper implements this, with a top-level acceptance of the new half with probability $\min\{1, n'/n\}$.

3.6 Step-size adaptation by dual averaging

PAPER Let H_t be a statistic with expectation $h(x) = \lim_{T\to\infty} \frac{1}{T} \sum_{t \le T} \mathbb{E}[H_t \mid x]$ for a tunable scalar x. The Robbins and Monro (1951) update $x_{t+1} = x_t - \eta_t H_t$ converges to a root of h if $\sum_t \eta_t = \infty$ and $\sum_t \eta_t^2 < \infty$, but its diminishing steps weigh early iterations too heavily. Dual averaging (Nesterov, 2009) is adapted to

$$x_{t+1} = \mu - \frac{\sqrt{t}}{\gamma} \frac{1}{t + t_0} \sum_{i=1}^{t} H_i, \qquad \bar{x}_{t+1} = \eta_t x_{t+1} + (1 - \eta_t)\, \bar{x}_t, \quad \eta_t = t^{-\kappa}. \tag{13}$$

- μ — Point towards which iterates are shrunk; set to $\log(10\varepsilon_1)$ so that larger step sizes, which cost less, are tried first.
- γ — Strength of shrinkage; 0.05 in all experiments.
- t_0 — Stabiliser of early iterations; 10 in all experiments.

κ Exponent of the averaging weight, in $(0.5, 1]$; 0.75 in all experiments.

The tuned variable is $x = \log \varepsilon$ and the statistic is $H_t = \delta - \alpha_t$, so a mean acceptance above the target δ yields negative H and a larger ε. For HMC α_t is the Metropolis acceptance probability of Equation (3). For NUTS, which has no single accept-reject step, it is

$$H_t^{\text{NUTS}} = \frac{1}{|\mathcal{B}_t^{\text{final}}|} \sum_{(\theta,r)\in\mathcal{B}_t^{\text{final}}} \min\left\{1, \frac{p(\theta, r)}{p(\theta_{t-1}, r_{t,0})}\right\}, \tag{14}$$

the average acceptance probability that HMC would assign to the states of the final doubling. The initial value ε_1 is found by repeatedly doubling or halving ε until a single leapfrog step has acceptance ratio crossing 0.5 (Algorithm 4).

EVIDENCE A numerical illustration shows the size of the first update. With $\delta = 0.65$, $t_0 = 10$, $\gamma = 0.05$ and a first iteration that is rejected ($\alpha_1 = 0$), Equation (13) gives $\bar{H}_1 = 0.65/11 = 0.059$ and $\log \varepsilon_2 = \mu - (1/0.05)(0.059) = \mu - 1.18$, so $\varepsilon_2 = 10\varepsilon_1 e^{-1.18} \approx 3.1\,\varepsilon_1$. Without t_0 the same rejection would give $\log \varepsilon_2 = \mu - 13$ and $\varepsilon_2 \approx 10\varepsilon_1\, e^{-13}$, a catastrophically small step. This is the reason for $t_0 > 0$, which the authors explain as avoiding extreme values that would waste computation. The role of $\kappa < 1$ is to let the averaged iterate forget the transient burn-in values faster than a uniform average ($\kappa = 1$) would.

3.7 Stochastic volatility target

PAPER The fourth benchmark is a stochastic volatility model for 3000 daily S&P 500 returns,

$$\tau, \nu, s_1 \sim \text{Exponential}(100), \qquad \log s_i \sim \mathcal{N}(\log s_{i-1}, \tau^{-1}), \qquad \frac{\log y_i - \log y_{i-1}}{s_i} \sim t_\nu. \tag{15}$$

y_i Index level on day i; $\log y_i - \log y_{i-1}$ is the daily log return.

s_i Latent scale (volatility) on day i, following a Gaussian random walk in logs with precision τ.

ν Degrees of freedom of the Student-t innovations.

The precision τ is integrated out analytically, which the authors report speeds mixing, giving a 3001-dimensional target

$$p(s, \nu \mid y) \propto e^{-0.01\nu} e^{-0.01 s_1} \left(\prod_{i=1}^{3000} t_\nu(s_i^{-1}(\log y_i - \log y_{i-1}))\right) \left(0.01 + \tfrac{1}{2}\sum_{i=2}^{3000} (\log s_i - \log s_{i-1})^2\right)^{-3001/2}. \tag{16}$$

INFERENCE The last factor can be checked. With rate 0.01 for τ and $T = 3000$ scales there are $T - 1 = 2999$ Gaussian increments, contributing $\tau^{2999/2} \exp\{-\frac{\tau}{2}\sum(\Delta \log s_i)^2\}$. Multiplying by $0.01 e^{-0.01\tau}$ and integrating over τ gives a Gamma integral proportional to $(0.01 + \frac{1}{2}\sum(\Delta \log s_i)^2)^{-(2999/2+1)} = (\ldots)^{-3001/2}$, matching Equation (16). The reading of $\text{Exponential}(100)$ as an exponential with mean 100 (rate 0.01) is consistent with the factors $e^{-0.01\nu}$ and $e^{-0.01 s_1}$.

4 Data and Empirical Strategy

4.1 Data

PAPER Four target distributions are used; the empirical evidence is a controlled comparison of samplers on fixed targets, not an observational study.

Table 1. **Targets used in the paper's experiments**

Target	Construction	Data	Dimension
Multivariate normal (MVN)	Zero mean, precision matrix A drawn once from a Wishart with identity scale and 250 degrees of freedom	Synthetic	250
Logistic regression (LR)	Intercept plus 24 standardised predictors, independent $\mathcal{N}(0, 100)$ priors	German credit data, UCI repository, 1000 customers	25
Hierarchical logistic regression (HLR)	As LR with all two-way interactions, prior variance σ^2 given an exponential prior with rate 0.01	Same	302
Stochastic volatility (SV)	Equation (16)	3000 daily S&P 500 returns (period not stated in the version read)	3001

Source: reproduced from the paper's section 4.1 (official). The HLR dimension (302) is derived by the reviewer from the paper's counts: 300 coefficients, the intercept and the variance term.

4.2 Empirical specification and identification

PAPER Each sampler runs 2000 iterations, adapting ε in the first 1000 and discarding those 1000 as burn-in. HMC is run at 10 logarithmically spaced target simulation lengths $\lambda \approx \varepsilon L$, the largest 40 times the smallest (successive ratio $40^{1/9} \approx 1.5$), with 8 values of δ between 0.25 and 0.95; NUTS at 15 values of δ in the same range; each cell with 10 random seeds. The total is $10 \times 8 \times 4 \times 10 = 3200$ HMC runs and $15 \times 4 \times 10 = 600$ NUTS runs. Efficiency is the minimum over dimensions of the ESS, computed for both the mean and the second central moment, divided by the number of gradient evaluations. The minimum is taken because a sampler must explore every direction. The second moment is included because resonance at certain εL can make HMC draws antithetic, giving a low-variance estimate of a mean and a high-variance estimate of a variance. The effective sample size is defined as

$$\mathrm{ESS}_{q,f}(\theta_{1:M}) = \frac{M}{1 + 2\sum_{s=1}^{M-1}\left(1 - \frac{s}{M}\right)\rho_s^f}, \qquad \widehat{\mathrm{ESS}} = \frac{M}{1 + 2\sum_{s=1}^{\hat{M}^f_{\text{cutoff}}}\left(1 - \frac{s}{M}\right)\hat{\rho}_s^f}, \tag{17}$$

ρ_s^f Lag-s autocorrelation of $f(\theta_t)$ under the chain.

$\hat{M}^f_{\text{cutoff}}$ First lag at which the estimated autocorrelation falls below 0.05.

$\hat{\rho}_s^f$ Autocorrelation estimate centred on a mean and variance taken from a separate 50,000-iteration NUTS run with $\delta = 0.5$.

Using external moments avoids the downward bias in estimated autocorrelation when a chain has not converged. The identifying variation is therefore purely algorithmic: the same target, fixed data, varying sampler and tuning, with seeds supplying the randomness.

4.3 Data limitations and biases

- Reference moments for HLR and SV come from a NUTS run, so the ESS of NUTS and HMC are both measured against a quantity produced by one of the competitors. For MVN the truth is analytic and this concern does not arise.
- The hyperparameters γ, t_0 and κ were chosen by hand on the SV model, which is also one of the evaluation targets (a mild form of tuning on the test set).
- Four targets, all log-concave or close to it apart from the SV scale structure, are a narrow base for a claim of general adequacy.
- The period of the S&P 500 sample is not given in the version read, which limits replication of the SV target.
- The metric ignores wall-clock time, which matters when gradient evaluation is not the only cost (the authors note inner products are negligible only except for very simple models).

CRITIQUE Equation (16) as printed has no factor $1/s_i$ from rescaling the Student-t density, and no Jacobian factor for working with s_i rather than $\log s_i$. The expression for the HLR target also carries a term $-\frac{N}{2}\log\sigma^2$ with $N = 1000$, whereas a Gaussian prior with common variance σ^2 over 301 coefficients would produce a factor of 301 (the dimension) rather than the sample size. Whether these are typographical or reflect a different parametrisation cannot be determined from the version read; they do not affect the algorithms but would matter for replication.

5 Results

5.1 Dual averaging reaches its target

PAPER Figure 3 of the paper plots realised against target acceptance statistics, for NUTS and for HMC at several λ. The authors report that dual averaging usually coerces h to δ, and does worse for SV, which needs longer to burn in, particularly for HMC with small δ where high rejection slows convergence. Figure 4 shows $\bar{\varepsilon}$ for NUTS with $\delta = 0.65$ roughly converging within a few hundred iterations except for SV.

EVIDENCE The result supports H2 under their metric but is shown graphically only; no numerical tolerance (for instance the median absolute gap between h and δ) is reported.

5.2 Trajectory lengths

PAPER Histograms of NUTS trajectory lengths (Figure 5) show that most lengths are integer powers of two, so the U-turn rule usually fires after a doubling is complete rather than at an inner subtree. This is desirable because an inner firing forces the whole last half-tree to be discarded. Trajectory length rises with δ, since a higher target gives a smaller ε, which needs more steps before doubling back.

5.3 Efficiency of NUTS against HMC

PAPER Figure 6 compares ESS per gradient evaluation. HMC peaks near $\delta = 0.65$; NUTS peaks near $\delta = 0.6$ and is flat for $\delta \in [0.45, 0.65]$. On the two logistic regressions NUTS is about as efficient as HMC; on MVN and SV, NUTS with $\delta = 0.6$ outperforms the best HMC by about a factor of three. The best HMC simulation length varies by about a factor of 100 across targets, from 17.62 for MVN to 0.17 for LR.

EVIDENCE The result is reported in the text as ratios read off Figure 6; the underlying numbers are not tabulated. Economic or statistical significance in the usual sense does not apply, since the quantity is a computational efficiency ratio on fixed targets, with 10 seeds per cell as the only replication. No dispersion across seeds is reported in the text. A factor of three in ESS per gradient means three times fewer gradient evaluations for the same Monte Carlo precision, so a sampling run that takes an hour with tuned HMC would take about twenty minutes, before counting the preliminary runs that HMC needs to find λ.

5.4 Qualitative comparison with random-walk Metropolis and Gibbs

PAPER On the 250-dimensional MVN, NUTS with $\delta = 0.5$ used about one million gradient and likelihood evaluations; random-walk Metropolis used a million iterations at its theoretically optimal acceptance rate of 0.234 (Gelman et al., 1996, as cited), and Gibbs used a million sweeps. Figure 7 shows that Metropolis has barely begun to explore the first two coordinates, Gibbs explores part of the space, and NUTS produces effectively independent draws.

CRITIQUE The comparison is visual, on two coordinates of 250, and the budget is matched in gradient evaluations, which favours the gradient-based sampler. The authors concede that Gibbs and Metropolis would work much better after a rotation of the parameter space for a normal target, but argue that finding the rotation costs $O(D^3)$ and that both then need $O(D^2)$ operations per effectively independent sample.

5.5 Reviewer simulation (generated, not from the paper)

EXTENSION To test the headline claim independently, a reviewer implementation of Algorithms 5 and 6 was run on a 50-dimensional zero-mean normal with a precision matrix $A = GG^\top/(2D)$ where G is a $D \times 2D$ standard normal matrix (a Wishart with $2D$ degrees of freedom, scaled), drawn once with a fixed seed. Its eigenvalues lie in $[0.103,\ 2.91]$, a condition number of about 28. The sampler settings follow the paper: 2000 iterations, 1000 adaptation iterations, $\gamma = 0.05$, $t_0 = 10$, $\kappa = 0.75$, 3 seeds. ESS

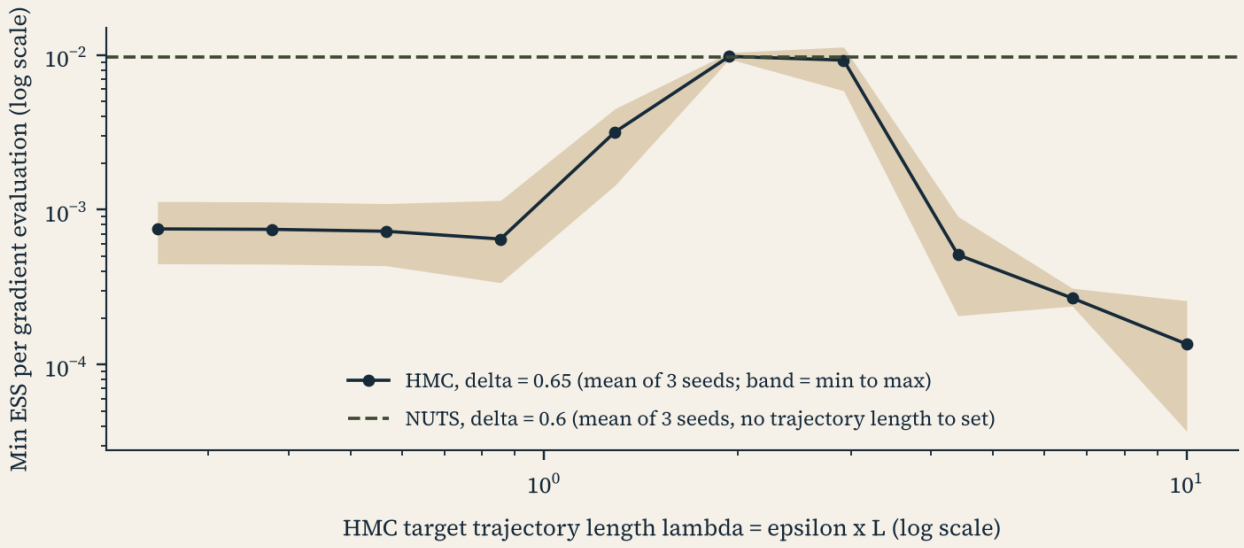
follows Equation (17) with the true mean and variance (known analytically), taking the minimum over the 50 coordinates and over the mean and second moment. This design differs from the paper in dimension, conditioning, number of seeds and ESS reference, so it checks the qualitative claim, not the factor of three.

Table 2. **Reviewer simulation: minimum ESS per gradient evaluation, 50-dimensional correlated normal**

Sampler and setting	Mean over 3 seeds	Minimum	Maximum
NUTS, $\delta = 0.4$	0.0091	0.0064	0.0113
NUTS, $\delta = 0.6$	0.0096	0.0095	0.0098
NUTS, $\delta = 0.8$	0.0065	0.0058	0.0077
HMC, $\lambda = 0.25$	0.00075	0.00045	0.00113
HMC, $\lambda = 1.29$	0.0032	0.0014	0.0045
HMC, $\lambda = 1.94$	0.0098	0.0094	0.0104
HMC, $\lambda = 2.92$	0.0092	0.0059	0.0112
HMC, $\lambda = 4.41$	0.00051	0.00021	0.00090
HMC, $\lambda = 10.0$	0.00014	0.00004	0.00026

Source: generated by the reviewer (not from the paper). HMC uses $\delta = 0.65$. Ten HMC lengths were run between 0.25 and 10; six are shown. A negative result for the three-fold claim is possible here because of the milder conditioning.

Figure 1. **Efficiency against HMC trajectory length on a 50-dimensional correlated normal (reviewer simulation)**



Source: generated by the reviewer. 2000 iterations, 1000 adaptation, 3 seeds per point. The horizontal line is NUTS at $\delta = 0.6$, which has no trajectory length to set.

EVIDENCE NUTS at $\delta = 0.6$ (0.0096) is statistically indistinguishable from the best HMC setting (0.0098 at $\lambda = 1.94$, 0.0092 at $\lambda = 2.92$) and about 1.0 times as efficient, not three times. The cost of a wrong guess is large, however: HMC at $\lambda = 0.25$ is 13 times less efficient than its best, at $\lambda = 4.41$ about 19 times, and at $\lambda = 10$ about 70 times. This reproduces the paper's central practical point, that HMC efficiency is sharply peaked in λ and NUTS removes the need to find the peak, while not reproducing the claim that NUTS beats the best HMC. The disagreement does not refute the paper: a Wishart matrix with as many degrees of freedom as dimensions, as used by the authors, has a condition number growing roughly like D^2 (an inference from random matrix theory, not checked here), so

their target is far harder than mine, and in harder problems adaptivity of trajectory length within an iteration plausibly gives more.

CRITIQUE A closer link to the paper would require its exact matrix, which is not given. The simulation also shows that NUTS needs on average about 20 gradient evaluations per iteration at $\delta = 0.6$, including the adaptation phase.

6 Robustness and Additional Tests

PAPER The paper has no section titled robustness. Its checks are embedded in the design: a grid of δ values, a grid of HMC trajectory lengths, 10 seeds per cell, four dissimilar targets, and a remark on the ESS truncation level.

Table 3. **Robustness tests: concern, test, result and assessment**

Concern	Test reported	Result	Does it resolve the concern?
Results depend on acceptance target	15 values of δ for NUTS, 8 for HMC	NUTS flat for $\delta \in [0.45, 0.65]$; HMC best near 0.65	Yes for the tested targets; the grid is coarse for HMC
HMC handicapped by poor λ	10 values of λ spanning a factor of 40	Best λ varies about 100-fold across targets	Mostly; the comparison uses the best λ per target, which is favourable to HMC and so conservative
Seed dependence	10 seeds per cell	Not summarised numerically in the text	Partly; dispersion is not reported
ESS estimator is biased by non-convergence	Reference mean and variance from a 50,000-iteration NUTS run	Authors say the approach gives more reliable intervals than CODA	Partly; reference comes from a competitor sampler
Cutoff of 0.05 in ESS is arbitrary	Authors vary it informally	Results not sensitive, per the authors	Unverified; no numbers shown
HMC anti-correlation (resonance) overstates ESS of means	Second central moment added to the metric	Minimum of the two used	Yes for this failure mode
Default adaptation constants may not generalise	Constants fixed across all targets	Work well on all four	Partly; chosen on SV, one of the test targets

Source: assembled by the reviewer from the paper's sections 3.2, 4 and Appendix A (assessment column is the reviewer's own).

7 Critical Evaluation

7.1 What the paper establishes

ESTABLISHED

The doubling construction with conditions C.1 to C.4 yields a transition kernel that, by the argument given, leaves the target invariant. A trajectory-length rule based on a displacement-momentum inner product is workable at the cost of a handful of inner products per leapfrog step. HMC efficiency per gradient evaluation is sharply sensitive to λ, and the best λ differs across problems by about two orders of magnitude (also seen in the reviewer simulation).

SUPPORTED BUT CONDITIONAL

That NUTS matches the best HMC on ESS per gradient holds on the four tested targets, with the metric as defined, and is reproduced qualitatively here; the three-fold gain depends on the strongly correlated targets and was not reproduced in a better-conditioned problem. The dual averaging scheme reaches its target acceptance statistic in most runs given a long enough adaptation phase.

SUGGESTIVE

That $\delta \approx 0.6$ is a good universal default for NUTS. The flat region $[0.45, 0.65]$ is shown on four targets; the reviewer simulation shows a clear drop at $\delta = 0.8$, so the default is plausible but not established.

NOT ESTABLISHED

Convergence of the adaptation scheme for NUTS in the formal sense. The monotonicity of h^{NUTS} in ε is assumed, not shown, and the statistic in Equation (14) is not a Metropolis acceptance probability. Performance on multimodal, funnel-shaped, heavy-tailed or constrained posteriors is also not established, and the claim that NUTS is suited to generic engines rests on the argument rather than tests.

7.2 Strengths

STRENGTHS RECOGNISED BY THE AUTHORS

- Removal of the most troublesome tuning parameter, which makes HMC usable in turnkey software.
- Efficiency at least as good as tuned HMC without preliminary runs.
- Handling of regions of zero posterior probability by stopping and sampling from the visited states, where HMC discards the whole trajectory.

STRENGTHS IDENTIFIED INDEPENDENTLY

- The ESS metric is conservative: the minimum over dimensions of the worse of the mean and the second moment penalises samplers that explore some directions poorly and exposes HMC resonance.
- Scale of the experimental grid (3800 runs) lets the paper compare NUTS against the best HMC, not an arbitrary HMC.
- Correctness is argued from explicit conditions rather than asserted from practical success, which makes the construction extensible (the later literature builds on the same slice-and-tree structure).
- The method degrades gracefully: the stopping rule on divergent energy (Equation (8)) prevents astronomically improbable states from entering the candidate set.

7.3 Weaknesses and critique

SINGLE MOST IMPORTANT WEAKNESS

Problem. The empirical validation uses four targets that are all smooth and effectively unimodal, with no funnel geometry, strong multimodality or heavy-tailed marginals, and efficiency is judged by a single metric. **Why it matters.** The justification for a turnkey sampler is robustness on exactly the models where users do not know the geometry, and hierarchical models with scale parameters can have posterior regions where a single global ε is wrong everywhere. **Potential bias.** Favourable selection of benchmarks overstates generality; the U-turn rule may stop too early in narrow necks or too late in wide regions where the step size is mismatched. **How it could be tested.** Run NUTS on Neal's funnel (Neal, 2003), an eight-schools type hierarchical model in centred and non-centred form (Papaspiliopoulos et al., 2007), and a bimodal mixture, recording divergences, the bias of the funnel scale marginal against known truth, and ESS of tail quantiles.

OTHER WEAKNESSES

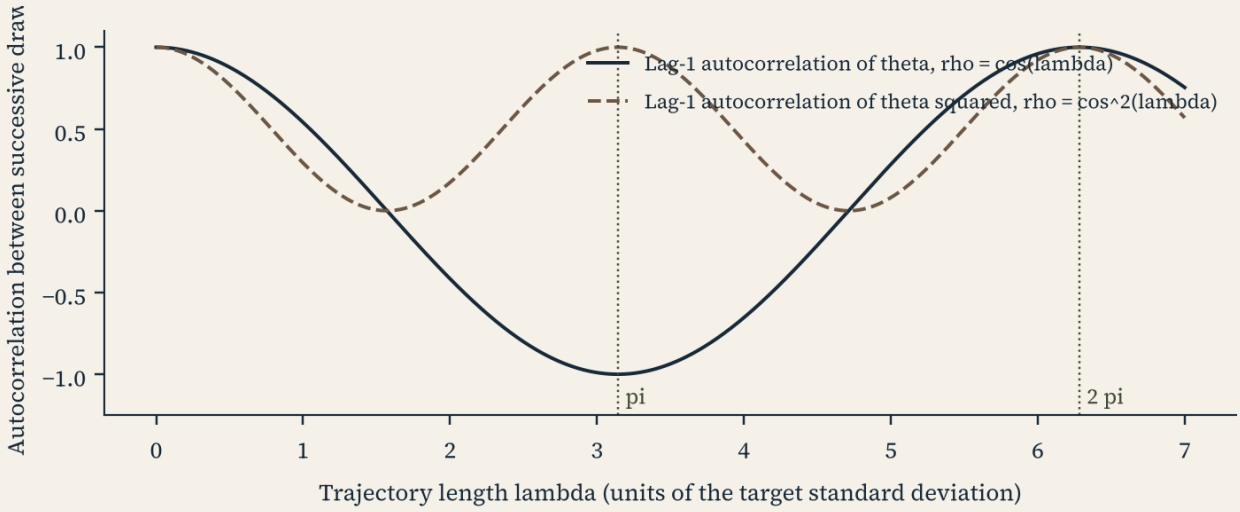
- Hand-tuned constants γ, t_0, κ and $\Delta_{\max}$ remain; the claim of no hand-tuning applies only to L and ε. Test: sensitivity grid over (γ, t_0, κ) on all targets.
- The adaptation is analysed with Robbins and Monro assumptions that do not obviously hold, because H_t^{NUTS} depends on the whole tree. Potential bias: the realised average acceptance can differ from δ without being flagged. Test: report the gap between realised and target statistics numerically across runs.
- No mass matrix. The identity metric implies the same ε in every direction, so with the strong scale differences found in real posteriors the step size is set by the stiffest direction. Test: compare with a diagonal mass matrix estimated from warm-up draws.
- The reference moments for HLR and SV come from a NUTS run. Test: use an independent long run from another sampler or a different seed family.

- Statistical reporting is graphical. Without tabulated means, standard errors across seeds, and the matrix used for MVN, the headline factor of three cannot be independently verified.
- Equation typos in the version read (see section on data limitations) reduce replicability.

8 Contradictions and Edge Cases

CRITIQUE **Resonance and periodic targets.** For a one-dimensional standard normal the exact dynamics are $\theta(\lambda) = \theta_0 \cos\lambda + r_0 \sin\lambda$. After an HMC move of length λ with $r_0 \sim \mathcal{N}(0,1)$, the new draw has correlation $\rho = \cos\lambda$ with the old one, and the squared value has correlation $\cos^2\lambda$ (since $\mathrm{Cov}(\theta^2, \theta'^2) = 2\cos^2\lambda$ and $\mathrm{Var}(\theta^2) = 2$). Hence, since the chain for the mean is exactly first-order autoregressive, the effective sample size per draw for the mean is $(1-\cos\lambda)/(1+\cos\lambda)$, equal to zero at $\lambda = 2\pi$ (the chain does not move) and larger than one for λ near π (the chain is antithetic), whereas for the second moment the same ratio with $\cos^2\lambda$ never exceeds one and is zero at both π and 2π. A fixed $\lambda = \pi$ therefore looks excellent for the mean and terrible for the variance, which is why the paper includes the second moment. NUTS avoids exact periodicity because its random doubling directions make the stopping time depend on (θ_0, r_0), and in the harmonic case the displacement criterion Equation (4) must fire within half a period (an inference from the orbit being an ellipse in (θ, r) space).

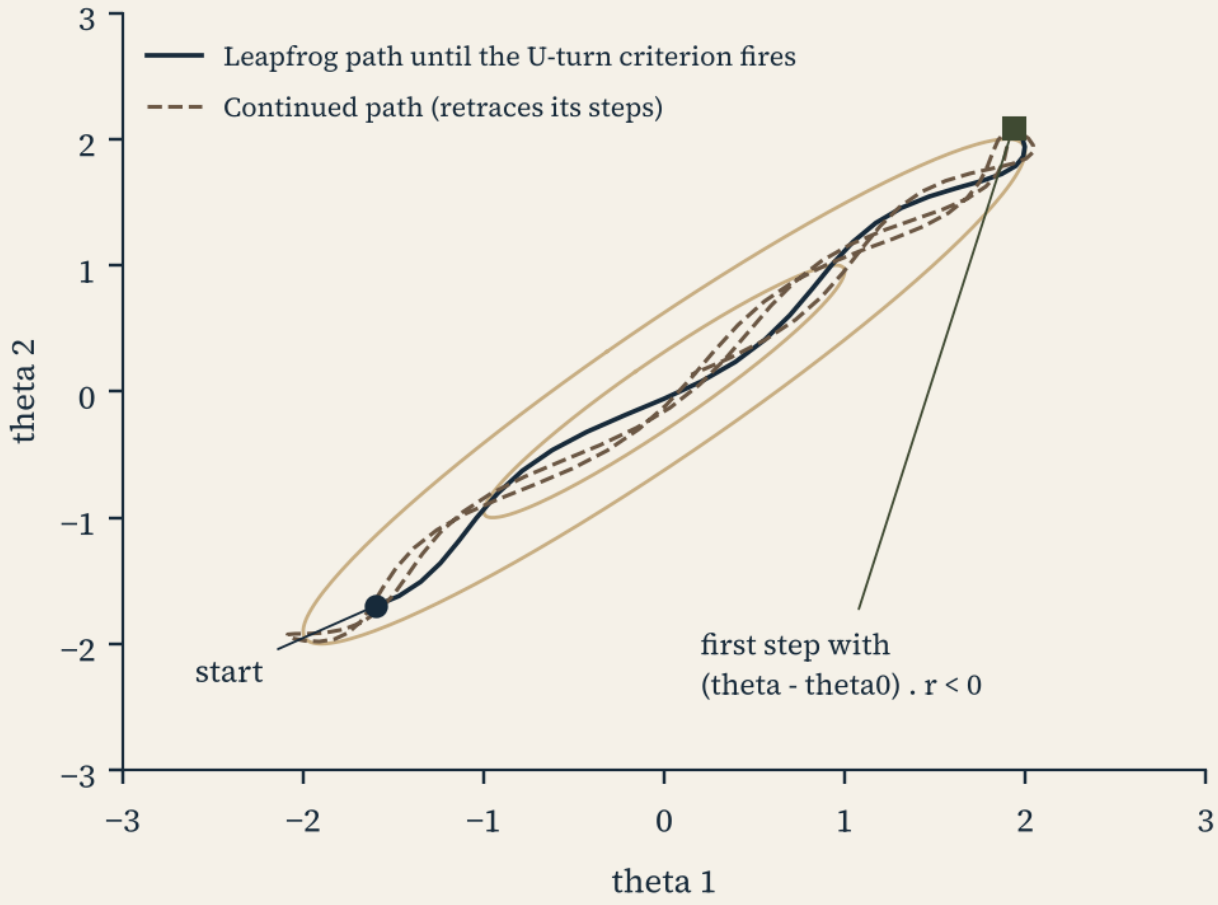
Figure 2. **Lag-1 autocorrelation of successive HMC draws for a standard normal target, as a function of trajectory length**



Source: computed analytically by the reviewer from the exact dynamics $\theta(\lambda) = \theta_0 \cos\lambda + r_0 \sin\lambda$. Dotted lines mark half and full period.

CRITIQUE **Illustration of the stopping rule.** In a strongly correlated two-dimensional normal the leapfrog path leaves the start, reaches the far edge of the ellipse and returns. The first step at which the inner product in Equation (4) turns negative marks the turning point; beyond it the path retraces ground it already visited.

Figure 3. **A leapfrog trajectory in a correlated normal (correlation 0.95), with the first U-turn marked**



Source: generated by the reviewer; step size 0.12, start $(-1.6, -1.7)$ and initial momentum $(1.1, 0.55)$; the ellipses are the one and two standard deviation contours. The criterion fires at step 30.

CRITIQUE **Other conditions under which the result could disappear.** First, when ε exceeds the stability limit $2/\sqrt{\lambda_{\max}(A)}$ derived above, the leapfrog map is unstable and energy diverges; Equation (8) stops the tree, but the acceptance statistic then collapses and dual averaging must recover. Second, the U-turn test uses the Euclidean dot product, which is correct only for the identity kinetic metric; with a mass matrix M the displacement-velocity product becomes $(\theta^+ - \theta^-) \cdot M^{-1} r$, so the paper's criterion cannot be used unchanged with a non-identity metric (an inference, in line with the authors' own remark that a mass matrix is a natural extension). Third, in a funnel the correct step size varies by orders of magnitude between neck and mouth, so with one global ε trajectories in the neck are inaccurate while those in the mouth are inefficiently short. Fourth, for discrete parameters or non-differentiable densities the gradient is undefined and NUTS does not apply, as the authors state. Fifth, when the target is so well conditioned that a single leapfrog step already gives near-independent draws, the extra bookkeeping of the tree has no efficiency to recover, consistent with the logistic regressions where NUTS merely matched HMC.

CRITIQUE **Scaling comparison.** Taking the stated costs per independent sample, D^2 for random-walk Metropolis against $D^{5/4}$ for HMC, the ratio is $D^{3/4}$: about 63 at $D = 250$ and about 405 at $D = 3001$. These are ratios of asymptotic rates without constants, so they indicate how the advantage grows with dimension, not absolute costs.

9 Further Literature

Table 4. **Literature connections**

Paper	Shared idea	Relationship	Why it matters
Duane et al. (1987)	Hybrid Monte Carlo with leapfrog dynamics	Preceded; origin of the method	Defines the algorithm that NUTS modifies
Neal (2011)	Review of HMC, including tuning of ε and L	Preceded; source of the tuning problem	States the heuristics NUTS replaces
Neal (2003)	Slice sampling with doubling and the funnel example	Preceded; supplies the doubling device	Basis for the tree and for the funnel stress test
Beskos et al. (2013)	Optimal HMC acceptance rate near 0.65 under strong assumptions	Preceded; justifies the HMC default ∂	Gives the theory behind the target ∂
Girolami and Calderhead (2011)	Position-dependent mass matrices	Alternative; addresses geometry, not path length	Complementary; combining with NUTS flagged by the authors
Betancourt (2017)	Conceptual and geometric account of HMC, with a refined NUTS in Stan	Extended later	Reports the refinements adopted in practice; the specifics should be checked against the text
Carpenter et al. (2017)	Stan, the system built on NUTS	Applied; realises the paper's stated aim	Shows the sampler as the default engine of a turnkey tool
Geyer (1992)	Autocorrelation truncation for ESS	Different methodology	Alternative to the 0.05 cutoff used here
Vehtari et al. (2021)	Rank-normalised, folded $\hat{R}$ and ESS for tails and bulk	Improved diagnostics	Addresses the mean-and-variance-only metric
Kim et al. (1998)	Likelihood inference for stochastic volatility via a mixture approximation	Different methodology for the SV target	A benchmark against which NUTS can be compared in finance
Papaspiliopoulos et al. (2007) and Betancourt and Girolami (2015)	Parametrisation of hierarchical models and its effect on HMC	Contradicts the generality of the claim	Shows geometry, not trajectory length, can dominate

Source: reviewer's assessment. Existence of the references not taken from the paper's own bibliography was checked through public bibliographic records; statements about Betancourt (2017) and Betancourt and Girolami (2015) rest on those records and general knowledge and were not verified against full texts.

10 Research Gaps

10.1 Paper-level gaps

- **Gap:** no convergence theory for the adaptive NUTS chain. *Why it matters:* adaptation that does not vanish properly can bias inference. *Existing limitation:* the guarantee is imported from stochastic approximation with an unverified monotonicity assumption. *Way forward:* prove or test monotonicity and the diminishing-adaptation property for H^{NUTS}.
- **Gap:** no hard targets. *Why it matters:* real finance models are hierarchical and heavy-tailed. *Existing limitation:* four benchmarks. *Way forward:* funnel, bimodal and constrained-parameter suites with divergence diagnostics.

10.2 Literature-level gaps

- **Gap:** comparison of gradient-based samplers with exact, structure-exploiting samplers for state-space models. *Why it matters:* for stochastic volatility, Gibbs samplers based on mixture approximations (Kim et al., 1998) are specialised and may be faster per unit time. *Existing limitation:* comparisons use gradient counts, not time. *Way forward:* wall-clock benchmarks on common data.
- **Gap:** adaptation of the metric together with the path length. *Why it matters:* with a fixed identity metric the step size is limited by the stiffest direction. *Existing limitation:* left to future work by the authors. *Way forward:* joint adaptation with convergence tests.

10.3 Methodological gaps

- **Gap:** the efficiency metric. *Why it matters:* means and second moments do not measure tail or quantile accuracy, which is what risk measures require. *Existing limitation:* the metric as defined. *Way forward:* quantile-based ESS and Monte Carlo error for tail probabilities.

11 Research Opportunities

Idea 1. NUTS against structure-exploiting samplers for stochastic volatility with leverage and jumps

RESEARCH QUESTION. On realistic stochastic volatility models for equity indices, does NUTS beat mixture-based Gibbs samplers on ESS per second for the volatility path and for the tail parameters?

MOTIVATION. Practitioners estimate volatility models daily; the paper's SV target omitted leverage and jumps.

HYPOTHESIS. NUTS wins when leverage correlation is strong and the Gaussian mixture approximation is poor; the specialised Gibbs sampler wins in the plain model.

ECONOMIC MECHANISM. Leverage couples return and volatility innovations, making the posterior of the latent path strongly curved.

LITERATURE CONNECTION. Kim et al. (1998); this paper; Vehtari et al. (2021).

DATA. Daily index returns from public sources, such as the S&P 500 from Yahoo Finance or Stooq, and the FRED series SP500 for a short check (availability must be confirmed at the time).

VARIABLES. Returns, latent log volatility, leverage correlation, jump intensity.

METHOD. Implement both samplers; compare bulk and tail ESS per second and posterior agreement.

IDENTIFICATION STRATEGY. Controlled simulation with known parameters, then real data; compare recovery of true parameters.

EXPECTED CONTRIBUTION. A finance-relevant benchmark absent from the paper.

MAIN RISK. Differences may reflect implementation quality more than algorithms.

Idea 2. Funnel failure in hierarchical return models

RESEARCH QUESTION. When do hierarchical priors on firm-level alphas or betas create funnel geometry that defeats a global step size, and how large is the resulting bias in the shrinkage parameter?

MOTIVATION. Bayesian cross-sectional asset pricing relies on hierarchical shrinkage.

HYPOTHESIS. With few observations per firm, centred parametrisations yield biased shrinkage and divergences; the non-centred form removes them.

ECONOMIC MECHANISM. The group scale and group effects are strongly coupled when data are weak.

LITERATURE CONNECTION. Neal (2003); Papaspiliopoulos et al. (2007); Betancourt and Girolami (2015).

DATA. Monthly stock returns for a US cross-section (from Kenneth French's data library for portfolios, or CRSP if accessible); availability to be checked.

VARIABLES. Excess returns, factor returns, firm alphas, group scale.

METHOD. Fit centred and non-centred models by NUTS; vary sample length and number of firms.

IDENTIFICATION STRATEGY. Simulation with known shrinkage scale; measure bias against sample size.

EXPECTED CONTRIBUTION. Practical guidance on parametrisation for Bayesian asset pricing.

MAIN RISK. The result may be known to practitioners and add little.

Idea 3. Tail-focused effective sample size for value at risk under Bayesian volatility models

RESEARCH QUESTION. Does a NUTS chain that has adequate bulk ESS give adequate Monte Carlo accuracy for 1 percent value at risk?

MOTIVATION. The paper's metric concerns means and variances, while risk management needs tails.

HYPOTHESIS. Tail ESS is a small fraction of bulk ESS for heavy-tailed ν posteriors.

ECONOMIC MECHANISM. Heavy-tailed marginals produce rare long excursions that the sampler visits slowly.

LITERATURE CONNECTION. Vehtari et al. (2021); Geyer (1992).

DATA. Daily index returns as in Idea 1.

VARIABLES. Posterior predictive 1 percent quantile, tail ESS, bulk ESS.

METHOD. Repeat long and short chains; estimate the Monte Carlo standard error of quantiles empirically.

IDENTIFICATION STRATEGY. Compare short-chain quantile dispersion against a very long reference chain.

EXPECTED CONTRIBUTION. A risk-management-oriented efficiency metric.

MAIN RISK. Dependence on the long reference run.

Idea 4. Joint adaptation of path length and diagonal metric

RESEARCH QUESTION. Does adapting a diagonal mass matrix during warm-up improve NUTS beyond what the U-turn rule achieves on posteriors with heterogeneous scales?

MOTIVATION. The paper uses the identity metric; the criterion changes with a metric.

HYPOTHESIS. Gains grow with the spread of marginal scales, and vanish for standardised problems.

ECONOMIC MECHANISM. Not applicable; this is a computational mechanism: stiffness in scale limits ε.

LITERATURE CONNECTION. Girolami and Calderhead (2011); this paper's discussion.

DATA. Synthetic Gaussians with controlled scale spread, then the paper's German credit logistic regression (UCI).

VARIABLES. Scale ratio, ESS per gradient, step size.

METHOD. Implement the metric-aware criterion $(\theta^{+} - \theta^{-}) \cdot M^{-1} r$ and test.

IDENTIFICATION STRATEGY. Controlled simulation over scale spread.

EXPECTED CONTRIBUTION. Quantifies an extension the authors left open.

MAIN RISK. Already settled in software practice, so the contribution may be incremental.

Idea 5. Reproducing the factor of three: dependence on conditioning

RESEARCH QUESTION. How does the NUTS to best-HMC efficiency ratio depend on the condition number of a Gaussian target?

MOTIVATION. The reviewer simulation found parity at condition number 28 while the paper reports a factor of three on a harder matrix.

HYPOTHESIS. The ratio rises with the condition number and approaches a constant above some threshold.

ECONOMIC MECHANISM. Not applicable; this is a computational question about stiffness.

LITERATURE CONNECTION. This paper; Beskos et al. (2013).

DATA. Synthetic Wishart targets with degrees of freedom from D to $10D$.

VARIABLES. Condition number, ESS per gradient, trajectory length.

METHOD. Extend the reviewer code across a grid of conditioning and dimension, more seeds.

IDENTIFICATION STRATEGY. Fully controlled simulation.

EXPECTED CONTRIBUTION. A clean replication note.

MAIN RISK. Narrow scope.

11.1 Ranking

Table 5. **Ranking of research ideas (scores from 1 to 5, assigned by the reviewer)**

Idea	Novelty	Feasibility	Data	Method	Importance	Contribution	Total
1. NUTS against Gibbs for SV	4	4	5	4	4	4	25
2. Funnel in hierarchical returns	4	4	4	4	4	4	24
3. Tail-focused ESS for VaR	4	4	5	4	4	4	25
4. Joint metric and path length	3	4	4	4	3	3	21
5. Conditioning replication	2	5	5	4	2	2	20

Source: reviewer's subjective scoring.

The top three are Ideas 1, 3 and 2. Ideas 1 and 3 share a dataset and a finance-relevant question that the paper left open, are feasible with public data and are decisive in outcome either way. Idea 2 ranks third because it addresses the most important weakness identified above in a setting that matters to asset pricing, with a slightly weaker data position because firm-level data may require access.

12 Learning Framework

12.1 Prerequisite concepts

- Markov chains, detailed balance and invariant distributions.
- The Metropolis-Hastings algorithm and why proposals must be reversible and volume preserving.
- Hamiltonian mechanics: position, momentum, energy conservation, symplectic integrators.
- Slice sampling and the doubling procedure.
- Stochastic approximation (Robbins and Monro) and convex dual averaging.
- Autocorrelation and effective sample size.

12.2 Core concepts learned

- Trajectory length is the key tuning parameter of HMC, and a reversible rule can choose it.
- A sampler can be made valid by constructing a symmetric candidate set and a kernel that is invariant on it.
- Averaging iterates with decaying weights tunes a scale parameter better than diminishing Robbins and Monro steps.

12.3 Important equations

Equations (2) and (3) for the integrator and the acceptance; (4) and (9) for the stopping rule; (7) and (10) for the validity argument; (13) for step-size adaptation; (17) for the metric.

12.4 Methods learned

Leapfrog integration, slice augmentation, recursive tree building with progressive sampling, dual averaging with shrinkage, and the use of an externally estimated reference mean for autocorrelation.

12.5 Questions to test understanding

COMPREHENSION

1. Why does HMC need a Metropolis step if the dynamics conserve energy?
2. What does Equation (4) measure, and why is stopping at its first sign change not valid?
3. What does the parameter δ control, and what is the effect of a large δ on trajectory length?

ANALYSIS

1. Why are the states added in the last doubling excluded when a stopping rule fires inside it?
2. Verify that the leapfrog map for a standard normal has unit determinant and find its stability limit.
3. Why does the term t_0 prevent extreme step sizes in the first iterations?

RESEARCH

1. How would the U-turn criterion change with a non-identity mass matrix, and is the invariance argument unchanged?
2. Design a benchmark that would make NUTS fail, and say what diagnostic would reveal the failure.
3. How would you estimate tail quantile accuracy from a NUTS chain?

12.6 Papers to read next

Neal (2011); Betancourt (2017); Girolami and Calderhead (2011); Papaspiliopoulos et al. (2007); Vehtari et al. (2021).

13 Conclusion

13.1 One-sentence thesis

Hamiltonian Monte Carlo can choose its own trajectory length by a reversible doubling rule that stops when the path turns back, and its step size by dual averaging, making it a turnkey sampler as efficient as a tuned one on the tested problems.

13.2 Three most important insights

- HMC efficiency is sharply peaked in the trajectory length, and the peak moves by about two orders of magnitude across problems.
- Validity of the adaptive trajectory comes from the symmetry of the candidate set, not from the stopping rule alone.
- The step size can be tuned to a target acceptance by averaging iterates in log space, with shrinkage and a stabiliser.

13.3 Strongest evidence

The large grid of runs in the paper showing NUTS at or above the best HMC on each target, together with the reviewer simulation reproducing the large penalty of a badly chosen fixed trajectory length.

13.4 Biggest weakness

The benchmarks do not include geometries, such as funnels and multimodality, on which a turnkey sampler is most at risk.

13.5 Most important research gap

Formal convergence of the adaptive scheme and its behaviour on hierarchical posteriors with variable curvature.

13.6 Best research opportunity

A finance-relevant benchmark of NUTS against specialised Gibbs samplers for stochastic volatility, reported with tail-focused diagnostics.

13.7 Three papers to read next

Neal (2011); Betancourt (2017); Vehtari et al. (2021).

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